1. Data Generation & Env Seeding

Seeding & Generaton (Inputs & Run Seeding)



CARLA--SUMO Environments & Process Setup

2D + virtual city with V2I edge links

Generate N = 30,000 Tasks

Mixed-weather shares:
- Clear 87%
- Light Rain 8%
- Heavy Rain 3%
- Dense Smog 2%

2. Policy Execution & Task Processing

Execution (Evaluation Loop) Loops Panel

- Local-only
- Edge-only
- Greedy
- Deadline-aware
- Energy-aware DRL
- Proposed

FOR policy (6 Candidate Policies)

Split-inference execution



Split-inference execution (Vehicle, network, edge MEC)

Execution Data Logging

- Weather
- Lateny
- Vehicle Energy
- System Energy
- Drops
- Violations
- Accuracy
- Variables
- Violations
- Accuracy
- Decisions

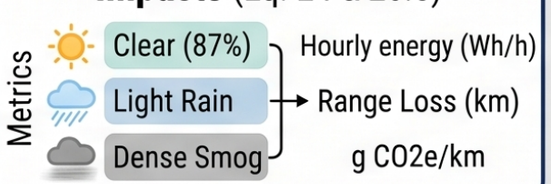
3. Metric Aggregation & Evaluation

Aggregation (Run-level Calculations)

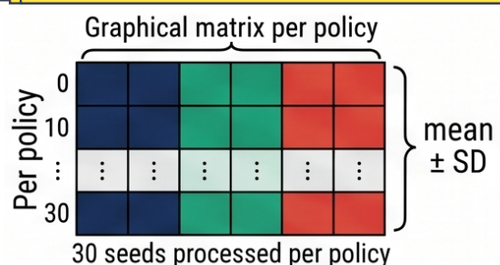
Compile Run-Level Metrics

- Mean/P95 Latency = mean/P95
- Deadline violations = dedlie violations
- Offload rate = offload rate
- Dropped tasks = dropped tasks
- Energy = mean(Wh/h)

Aggregate Weather & Vehicle Impacts (Eq. 24 & 26.5)



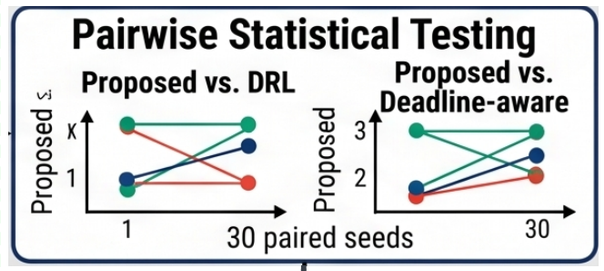
Accumulate Statistics



4. Statistical Testing & Significance

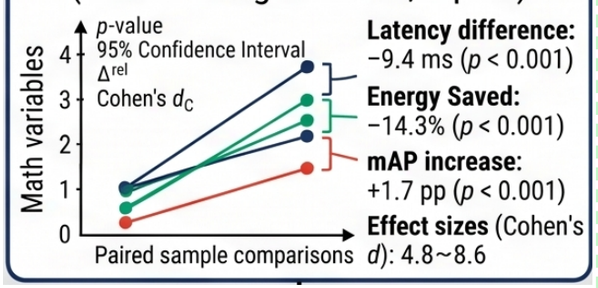
Statistical Testing

(Comparison & Verification)



Statistical Engine

Statistical Engine (Wilcoxon Signed-Rank, Eq. 24)



All Metrics Tabulated With Statistical Statistical Confidence